When Memory Metrics Hide Memory Policies

Same-Stream Evaluation and Policy-Facing Lookup Contracts for Persistent-Agent Memory

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Abstract

Persistent-agent memory systems can score well while hiding whether the underlying memory policy can govern, query, and repair the right evidence. We present a preregistered evaluation discipline for persistent-agent memory and demonstrate it on a consolidation-queue (CQ) case study. The discipline does four things. It runs every memory policy on the same candidate stream through the same substrate, and gates noisy-mode comparisons on extraction quality. It uses a formal policy-facing lookup-contract (PFLC) diagnostic to show that common clustering, retrieval, and answer-accuracy metrics can all pass while exact policy-query lookup fails. It transfers to an external benchmark (LongMemEval) under controls that preserve negative results rather than rescuing them. Finally, it pairs each interface repair with an independent control that reproduces the failure on a winning baseline.

In the case study, the discipline detects CQ's contradiction-handling survival on internal noisy extraction. On LongMemEval, it surfaces a stable external-transfer loss on evidence completeness, isolates that loss as a readout-cardinality interface gap, and closes the gap with a preregistered repair corroborated by a capped-Reflection control. These results support the evaluation discipline rather than a broad generalization claim for CQ.

1 Introduction

An agent helping a user across months learns small facts incrementally. In March, the user mentions they prefer espresso; in August, they are cutting caffeine and want tea in the morning. A naive memory system overwrites the older preference; a less naive one stores both as parallel facts. Neither resolves the conflict when the user later asks what to order. The policy question for persistent memory is whether the system can keep both claims, scope them, contest one with the other, and demote a durable claim when the evidence shifts.

Persistent-agent memory systems are usually judged by whether they recall facts, retrieve useful context, or answer held-out questions. Those metrics are important, but they can hide the policy question that matters for memory governance: did the system succeed because its memory policy handled contradiction, scope, source independence, and reversibility, or because it received easier inputs, richer private state, or a metric that did not inspect the policy-facing handle being queried?

This report argues for a stricter evaluation posture. Memory-policy comparisons should hold the upstream candidate stream fixed, use the same storage substrate for the policies being compared, separate oracle-mode policy claims from noisy-pipeline claims, and preserve artifacts that make failure cases inspectable. The running case study is a lightweight consolidation queue policy (CQ),[1] compared with immediate-write Reflection, a Mem0-style write-only baseline

[1] A non-archival design sketch is available at The Consolidation Queue: How Persistent AI Earns Durable Change.

(`Mem0WritePolicyLite`, not the production Mem0 system), and ablated variants. We use CQ as a case-study policy for testing which memory-policy distinctions survive the interfaces around them.

Concretely, CQ is a staged-write policy. Incoming candidate memories enter a queue with scope, provenance, support, contradiction, and lifecycle metadata. High-scoring candidates promote to durable memory; lower-scoring but usable candidates can remain pending and still answer scoped lookups. Contradictory evidence contests earlier candidates and can demote active durable memories, while narrower pending overrides can shadow a wider durable claim without globally overwriting it. In the opening example, CQ keeps both the espresso preference and the morning-tea preference as scoped claims: the narrower morning-tea claim shadows the broader espresso claim at read time, and either can be contested or demoted when stronger evidence arrives.

Internally, under a preregistered noisy policy comparison, CQ's contradiction-edge-driven contestation and demotion mechanism remains visible on forced contradiction and partially visible on preference drift. Other rows stay recorded as nulls or policy-facing lookup-contract failures. Externally, a controlled LongMemEval v1 transfer probe produces a stable negative result for base CQ on evidence completeness. A later preregistered follow-up repairs that specific failure by exposing all eligible same-slot pending evidence at answer time, while a cardinality-capped Reflection control reproduces the original loss. The paper preserves the failure and the repair as separate evidence.

The paper collapses the repository's detailed evidence ledger into four headline claims.

Claim 1: Fair-stream memory-policy evaluation. The benchmark fixes the same upstream `CandidateUpdate` stream and the same scoped memory substrate across policies, then gates noisy-mode policy claims on component quality. This removes asymmetric inputs, asymmetric storage, and hidden extractor advantages from the policy comparison. It also records lifecycle traces, manifests, failure examples, and replay checks rather than reducing the experiment to a single score.

Claim 2: Policy-facing lookup contracts. PFLC diagnostics name a class of metric blind spots. Proposition 1 proves a cluster-partition/lookup-contract gap: a relabeling can preserve the gold partition and therefore score 1.00 under partition metrics while failing exact policy-query set membership. Constructive lemmas extend the same pattern to retrieval-at-k and answer-accuracy metrics. The lookup diagnostic instantiates this class on canonical-id queries within the CQ benchmark; the field-level PFLC survey divides into an artifact-blocked majority (no per-question evidence ids exposed) and a published-output replayable subset (AMB LoCoMo) that decomposes aggregate answer accuracy into evidence exposure, rank, and answer-generation residuals.

Claim 3: Controlled external transfer with preserved negative results. The LongMemEval workstream builds a redacted same-stream adapter, dual-path annotation protocol, hidden-answer verifier, judge-calibration gate, PFLC scorer, and contract-sensitivity audit before policy execution. The result is stable and negative for base CQ on `all_hit_at_50`. CQ retrieves one gold evidence session in every case, tying eager baselines on `any_hit_at_50`; the missing second evidence session drives the `all_hit_at_50` loss.

Claim 4: Cardinality-controlled mechanism repair. The follow-up preserves the LongMemEval transfer probe as a negative base-CQ result and tests a separately preregistered interface hypothesis: plural pending readout should close the evidence-completeness gap, and capping a winning eager baseline to one returned candidate should reproduce the loss. The follow-up results match both predictions. The result is a mechanism-local repair pattern for this interface gap; broader LongMemEval and natural-language QA generalization remain outside its scope.

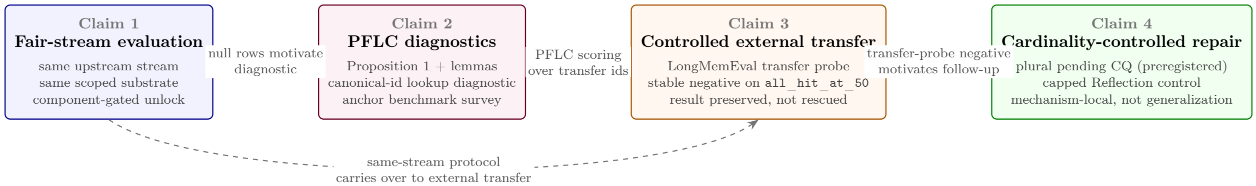

Figure 1: Claim dependency graph. Claim 1 (fair-stream evaluation) is the methodological foundation; the noisy same-stream policy comparison's null rows motivated Claim 2 (PFLC diagnostics), which then supplied the policy-facing scoring used by Claim 3 (controlled external transfer). Claim 1's same-stream protocol also carries over directly to Claim 3 (dashed edge). Claim 3's stable negative transfer-probe result motivated the preregistered interface follow-up in Claim 4 (cardinality-controlled repair). The contribution is the chain, not any individual claim.

All empirical claims in this draft cite committed artifacts. Where a result is diagnostic-only, descriptive-only, or blocked by missing artifacts, the paper keeps that boundary in the main text rather than relegating it to a caveat.

2 Related Work

External memory benchmarks. LongMemEval evaluates long-term chat memory across information extraction, multi-session reasoning, knowledge updates, temporal reasoning, and abstention [11]. Here, the released v1 oracle split becomes a controlled-pilot anchor only after a preregistered adapter creates a redacted same candidate stream for all local policies. The resulting PFLC target is dialog-evidence-id completeness over the v1 evidence-only oracle split, so the result should be read as evidence exposure and completeness under a fixed policy surface, with distractor-heavy retrieval quality over LongMemEval-S or LongMemEval-V2 left to future work. LoCoMo is the other key long-conversation benchmark used in the PFLC survey: it collects very long multimodal conversations and evaluates question answering, event summarization, and multimodal dialogue generation [6].

MemoryAgentBench and MemBench are important external-evaluation references because they stress incremental interaction, long-range memory, factual memory, and selective forgetting [5, 10]. They occupy the benchmark lane, separate from this report's fair-policy-comparison lane, because standard benchmark outputs need not enforce same-candidate-stream and same-substrate constraints. The PFLC survey treats their released artifacts as a diagnostic question: do standard outputs expose the per-question identifiers needed to test whether a memory policy can query the right handle?

The answer splits. Most surveyed released outputs are artifact-blocked, but published Agent Memory Benchmark (AMB) LoCoMo runs (Hindsight, hybrid-search, Cognee) inject context with recoverable LoCoMo `dia_id`s and admit a published-output PFLC replay, which decomposes aggregate answer accuracy into evidence exposure, rank, and answer-generation residuals (Section 4.4).

Belief revision and truth maintenance. CQ's contestation and demotion behavior is adjacent to classical truth maintenance and belief-revision work, which tracked reasons for beliefs, assumption sets, and revision under inconsistent or changing evidence [4, 3, 1]. This paper does not claim a new general belief-revision logic. It uses that lineage as a local memory-policy pattern for LLM agents, adding scoped candidate streams, queryable pending state, and same-stream comparison against eager-write policies.

Memory systems and write policies. Mem0 proposes a production-oriented memory architecture with dynamic memory extraction, update, and retrieval, including a graph variant for relational memory [2]. A-MEM emphasizes dynamic indexing, memory linking, and memory evolution [12]. MemGPT/Letta frames stateful agents through virtual context management and self-editing memory tiers [8]. Zep/Graphiti and Cognee represent graph-oriented memory and retrieval stacks that motivate the related-system lane without defining this report's fair policy-comparison surface [9, 7]. These systems motivate comparison targets and terminology. The local `Mem0WritePolicyLite` baseline is a narrow ADD/UPDATE/NOOP write-time policy over the same candidate stream and substrate as CQ and `ReflectionEagerWriteLite`; that narrowness keeps the policy comparison inspectable. `Mem0WritePolicyLite` is *not* an end-to-end reproduction of the Mem0 system, and our results do not characterize production Mem0; we use this paper-facing label (renamed from the repo identifier `Mem0Lite`) to make that boundary unambiguous to readers.

Preregistration and audit discipline. The methodological stance is closer to a preregistered evaluation audit than to a recall leaderboard. The project locks prediction grids, outcome buckets, adapter pins, judge gates, and follow-up boundaries before running each major comparison. It also records aborts as evidence. The canonical-id resolution audit, for example, refused to emit a verdict twice when locked run JSON payloads could not be reproduced. That posture prevents a methodology repair from becoming a post-hoc result rescue.

3 Method

3.1 Notation and Evaluation Stages

The core unit is a `CandidateUpdate`: a proposed memory item with claim content, scope key, provenance, evidence links, and lifecycle state. A *durable* memory is committed state. A *pending* candidate is not durable, but CQ can still expose it to scoped lookup while it remains reversible. A *same-stream* comparison gives every policy the same ordered `CandidateUpdate` stream; a *same-substrate* comparison stores every policy's state through the same scoped memory interface.

PFLC means policy-facing lookup contract: the identifier or handle a memory policy must expose so a query can be evaluated against the intended evidence. The *canonical-id lookup diagnostic* (repo label: QR-canon) is this paper's canonical-id PFLC instance. It reuses the set-membership metric and alias function from the earlier *canonical-id resolution audit* (repo: CQR), but it is a small committed diagnostic table rather than a replay of the large noisy same-stream policy comparison run payloads.

This paper uses descriptive names in body text; repository labels appear once in parentheticals to keep the artifact trail navigable. The evaluation steps are: the *prompt-regression guard* (repo: Phase A) runs before noisy component scoring; the *noisy extractor-quality gate* (repo: Phase 3) scores local extractor outputs and licenses downstream policy comparison; the *noisy same-stream policy comparison* (repo: Phase 4) runs after that gate passes; and the *cardinality-repair follow-up* (repo: Phase Y) addresses a specific external-transfer interface gap. Preregistered interpretation labels are: a planned gate or thesis condition fired (repo: Bucket A), a completed mixed or partial result (repo: Bucket B), and an abort or surprise-lane boundary that prevents the stronger claim (repo: Bucket D). The 7B and 32B labels denote local extractor model cells, not memory-policy sizes.

3.2 Benchmark Objects

The benchmark centers on scoped candidate updates and a shared memory substrate. Candidate updates carry claim content, scope, provenance, evidence relations, and lifecycle state. All policies receive the same upstream observations; they differ in when they promote, demote, contest, or expose these candidates. The load-bearing invariant is that CQ and the immediate-write baselines compare memory policy over the same inputs.

The principal policies are `ConsolidationQueueLite`, `ReflectionEagerWriteLite`, `Mem0WritePolicyLite`, `NaiveEagerWriteLite`, `NoMemoryLite`, and CQ ablations that remove contestation/demotion, pending lookup, source-independence gating, or wider-scope pending override. Two follow-up policies are explicitly outside the original headline set: `CQDatedContestation`, an oracle-mode dated-evidence repair, and `CQPendingMultiEvidence`, the LongMemEval pending-readout repair. The follow-up also includes `ReflectionEagerWriteCardinalityCapped` as a diagnostic control.

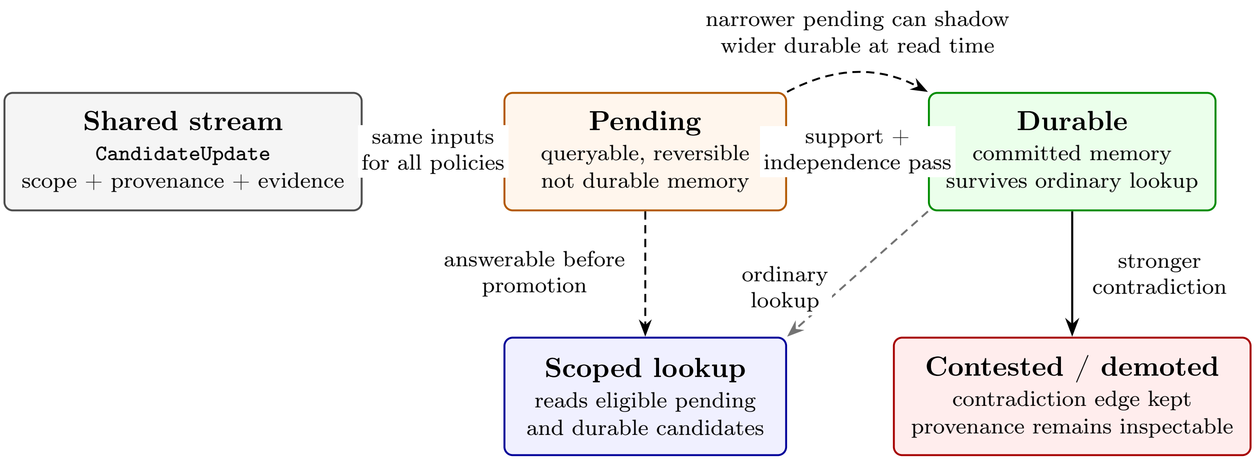

Figure 2: CQ lifecycle. Candidate updates come from the same upstream stream used by every policy in the comparison. CQ differs from eager writing by allowing a candidate to stay pending while still being queryable, by promoting only when support is strong enough, and by preserving contradiction edges so an active durable memory can be contested or demoted. A narrower pending candidate may shadow a wider durable claim for scoped lookup without globally overwriting it.

CQ trace example. Suppose a candidate says the user prefers espresso under the coffee-preference scope. With one source and moderate support, CQ can keep the candidate pending: it is answerable to scoped coffee queries, but it has not become durable memory. Later, a narrower morning-coffee candidate says the user prefers tea in the morning and carries stronger support plus a contradiction edge. CQ records the conflict, lets the narrower pending candidate shadow the broader coffee claim for morning lookups, and, if an older espresso claim had already promoted, demotes that durable record into contested state instead of deleting it. The record's source history remains inspectable. The eager baseline receives the same candidate stream, but writes immediately.

3.3 Oracle Stage

The oracle stage tests policy behavior when candidate observations, scope keys, contradiction/support relations, and source metadata are known. It includes forced contradiction, scope contamination, preference drift, useful pending memory, false corroboration, memory poisoning, mechanism-diverse frozen held-out scenarios, adversarial upstream noise, and the evidence-conflict spectrum. This stage establishes policy behavior under controlled evidence before extraction quality enters the experiment.

3.4 Noisy Stage

The noisy stage renders transcripts, scores local extractor outputs against component labels, converts accepted predictions into shared candidate streams, and only then runs policies. The component gate has layered checks: aggregate confidence-bound gates, per-family observed gates, frozen mechanism-diverse sentinels, prompt-regression checks, deterministic replay, and exact adapter pin validation. The 7B anchor passed aggregate gates but stayed locked because it had concentrated scenario errors and family/sentinel failures. The preregistered 32B cells cleared the gate and licensed the separate noisy same-stream policy comparison.

3.5 Attribution Stage

Policy results require separate attribution. The mixed-support mechanism audit reads policy-comparison artifacts, component-eval artifacts, ablation patterns, exact policy-query canonical-id alignment, and representative traces. It attributes forced contradiction cleanly to contradiction-edge preservation and contestation/demotion. Preference drift is partial survival: the policy delta remains, but absolute CQ performance drops relative to oracle and the 32B artifacts show residual drift. Other rows remain null, descriptive, or policy-facing lookup-contract failures unless an artifact directly supports a stronger claim.

3.6 Policy-Facing Lookup Contracts

PFLC is operational rather than a metric nicety because the governance operations a persistent-agent memory must support all require identifier-bearing state, not just relevant content. The evaluated operations in this paper are contradiction handling, contestation, demotion, provenance preservation, and scoped override: each must address a specific prior claim by handle, not by paraphrase or similarity. Cluster-partition metrics cannot test this because they are invariant to relabeling; retrieval-at-k metrics cannot test it because they score content overlap, not handle identity; answer-accuracy metrics cannot test it because a correct answer can be produced from non-canonical state. Production deletion, right-to-be-forgotten, and enterprise compliance are motivating examples of why a queryable, identifier-bearing memory state matters in deployed systems; this paper does not evaluate them.

PFLC diagnostics ask whether the system exposes the identifier or handle that a memory policy actually needs to query. In plain terms, clustering metrics treat labels as interchangeable, so a system that perfectly groups items into clusters $\{A, B, C\}$ scores 1.00 even if it called them $\{1, 2, 3\}$ internally; a governance query that asks "is item X in slot B?" fails when the system has slot 2 instead. The formal core makes this precise in three parts. Proposition 1 shows that for non-singleton label alphabets and cluster-partition metrics invariant to injective relabeling, the predicted partition can equal the gold partition while set-membership PFLC is zero for a queried gold id. Lemma 1 constructs the same gap for retrieval-at-k metrics: retrieved content can be relevant while the policy-facing evidence id is missing. Lemma 2 constructs it for answer-accuracy metrics: a generated answer can be correct while the system fails to expose the handle needed for memory governance.

The lookup diagnostic instantiates this PFLC class on canonical-id queries within the internal audit. It reuses the locked resolution audit's Section A set-membership metric and alias function, reads

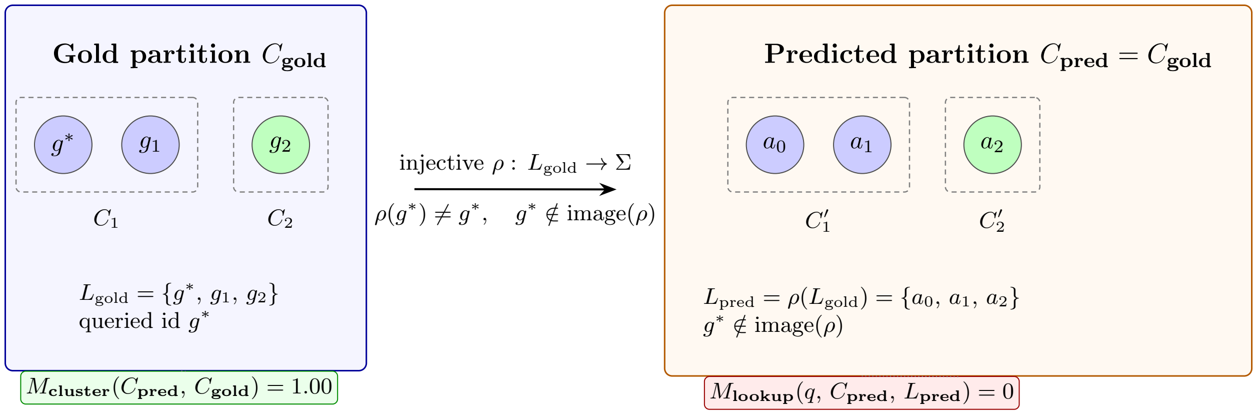

Figure 3: PFLC formal gap (Proposition 1, worked example). The predicted partition is identical to the gold partition, so any cluster-partition metric (M_{cluster}: B-cubed F1, ARI, NMI, V-measure, pairwise cluster F1) scores 1.00. The injective relabeling ρ maps L_{gold} into Σ with image disjoint from $\{g^*\}$, so set-membership PFLC for the queried id g^* is 0. The two outcomes depend on disjoint properties of the same predicted state: partition structure versus label image.

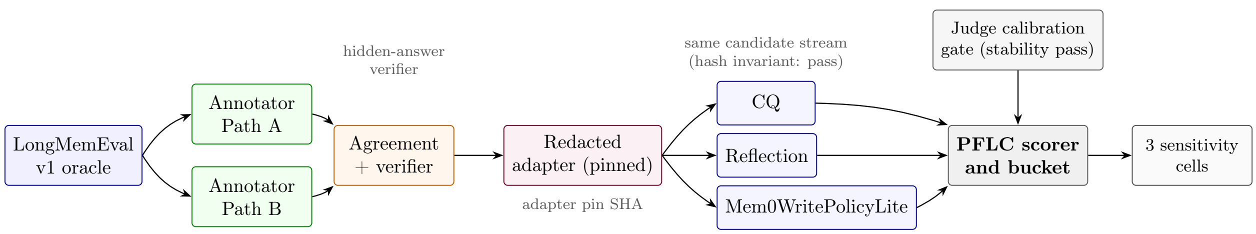

Figure 4: Controlled LongMemEval externalization. The v1 oracle source is annotated by two independent paths whose agreement is verified before any policy sees data. The redacted adapter strips the gold answer, pins a SHA-locked candidate stream, and feeds the same stream to every policy. The judge calibration gate must clear stability before scoring runs; the PFLC scorer reads policy outputs across three contract-sensitivity cells.

a committed per-question source table, and reports Wilson intervals plus a joint B-cubed/lookup table. It is diagnostic and leaves the noisy same-stream policy comparison verdict unchanged.

3.7 Controlled External Transfer

The LongMemEval workstream adapts an external benchmark into a controlled same-stream pilot. The protocol selects LongMemEval v1 as the primary anchor, freezes dual-path annotations, verifies the hidden-answer boundary, pins the adapter, calibrates a local judge, and uses PFLC scoring over answer-session ids. The v1 oracle split is evidence-only, so the PFLC metric measures whether policies preserve and expose gold evidence sessions they received, not whether they search through distractor haystacks.

The LongMemEval transfer probe uses `all_hit_at_50` as the headline metric and preserves `answer_correct` as diagnostic-only because policy answers are structured memory traces. The follow-up keeps the same adapter, judge, and headline metric but changes the policy-facing readout interface under a new preregistration: CQ exposes all eligible same-slot pending candidates, and capped Reflection returns only one durable candidate id at answer time.

4 Results

Table 1 maps the paper's four claims to their principal evidence anchors. The full artifact index is in Appendix D; this section emphasizes each result's supported scope.

Table 1: Claim-level evidence ledger.

Claim	Evidence	Readout	Boundary
Fair-stream evaluation	Frozen oracle sweep; noisy extractor-quality gate; noisy same-stream policy comparison	All 108 frozen oracle deltas matched. The 7B gate blocked a false unlock. The 32B-gated comparison reaches mixed/partial support.	Mechanism-local noisy support.
PFLC diagnostics	Lookup diagnostic; PFLC proposition and field survey	Five of seven comparison rows pass the B-cubed floor while exact canonical-id match is below 5 percent; formal and constructive counterexamples cover clustering, retrieval, and answer metrics.	Diagnostic PFLC; resolution audit verdict unchanged.
Controlled external transfer	LongMemEval transfer probe	Methodology gates pass, but CQ minus Reflection on `all_hit_at_50` is -1.0 in all three cells.	Stable base-CQ transfer loss.
Cardinality-controlled repair	Pending multi-evidence follow-up	`CQPendingMultiEvidence` moves base CQ from 0/71 to 71/71 on the primary cell; capped Reflection falls from 71/71 to 0/71.	Interface repair; external generalization out of scope.

4.1 Oracle Results

The frozen mechanism-diverse oracle sweep matched all preregistered deltas. CQ and `Mem0WritePolicyLite` tied on aggregate answer correctness, false assertion, poison promotion, clean durable displacement, and leakage. CQ's narrower advantage was lower premature promotion, especially on false-corroboration mixed-source rows. The result supports staged promotion as a way to reduce premature durable commitment under oracle inputs; broad answer-quality superiority over `Mem0WritePolicyLite` remains outside the supported claim.

The adversarial upstream-noise sweep is similarly mixed. CQ wins retraction, witness conflict, scope narrowing, and pending competition, but loses temporal skew. The post-hoc `CQDatedContestation` follow-up repairs base CQ's temporal-skew abstention into a correct answer from fresher dated evidence and preserves the four original wins. The CQ-vs-Reflection win criterion remains unmet because Reflection also lands at the correctness ceiling.

4.2 Noisy Component Gate and Policy Comparison

The component gate demonstrates why aggregate extractor metrics are insufficient. The locked 7B row passed prompt-regression, determinism, aggregate confidence-bound gates, and aggregate observed gates. It still stayed locked because it had 45 primary scenario errors, eight per-family observed failures, and three frozen-sentinel observed failures. An aggregate-only rule would have authorized a false noisy policy comparison.

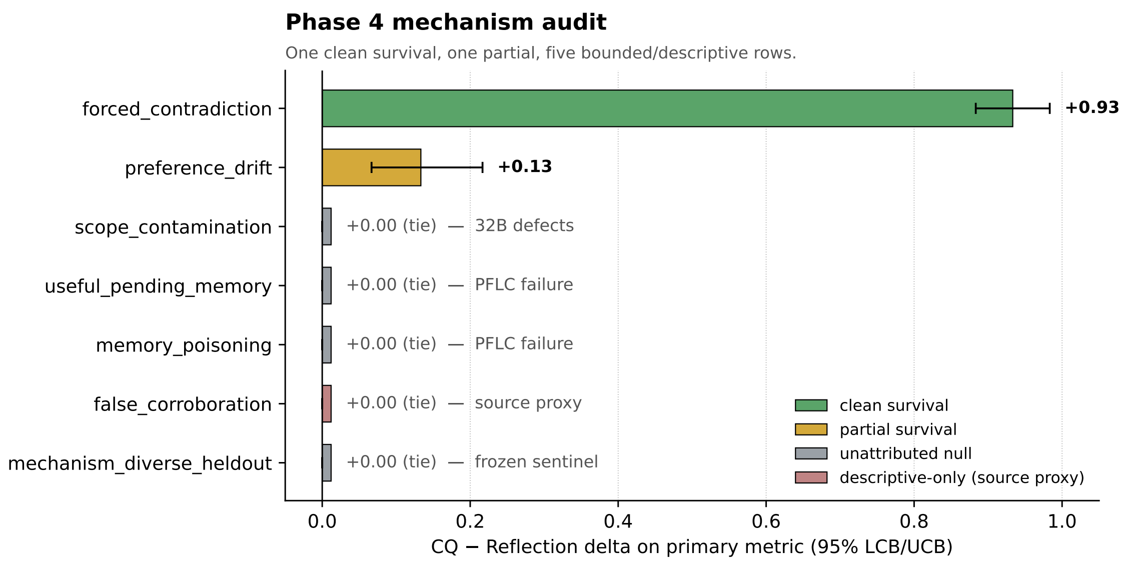

Figure 5: Noisy same-stream policy comparison mechanism audit summary. Only forced contradiction is a clean survival row; preference drift is partial, and other rows remain bounded, diagnostic, or descriptive evidence (component defects, lookup-contract failures, the descriptive-only event-source proxy for false corroboration, or the frozen-sentinel held-out tie).

The preregistered 32B rows cleared the gate and licensed the noisy same-stream policy comparison. The comparison reaches mixed/partial support: CQ wins against `ReflectionEagerWriteLite` on forced contradiction (delta $+0.93$, lower confidence bound $+0.88$) and preference drift (delta $+0.13$, lower confidence bound $+0.07$), records no countable directional losses, ties `Mem0WritePolicyLite` on frozen sentinel primary metrics, and has no replicate contradictions. The mechanism audit narrows this: forced contradiction is the clean mechanism-survival row, preference drift is partial survival, and the remaining rows stay null, descriptive, or bounded by component evidence.

4.3 Lookup Diagnostic and PFLC

The lookup diagnostic adds a second diagnostic layer to several tied rows. On `useful_pending_memory`, `memory_poisoning`, `false_corroboration`, and the frozen sentinel, B-cubed F1 is at or near 1.00 while exact policy-query canonical-id agreement is 0.00. The locked alias-normalized variant also stays at zero on those rows. This does not replace the dominant mechanism-audit labels in Figure 5: the `false_corroboration` row remains descriptive-only source-proxy evidence, and the mechanism-diverse row remains the frozen sentinel aggregate. It means the extractor can look cluster-correct while the policy-facing lookup contract fails. The policy verdict does not change.

The field-level PFLC work generalizes this observation. Proposition 1 and Lemmas 1–2 show that clustering, retrieval, and answer-accuracy metrics are insufficient evidence of policy-facing lookup success. The four-anchor survey over LongMemEval, Mem0/LoCoMo, MemoryAgentBench, and MemBench splits on artifact accessibility: most released baseline outputs are artifact-blocked, but a subset of published Agent Memory Benchmark (AMB) LoCoMo runs expose recoverable per-question evidence ids and support a published-output PFLC replay. The split is summarized below; benchmark-targeted synthetic counterexample rows ship alongside.

4.4 Released-Artifact PFLC: Blocked vs Replayable

Across the surveyed memory-benchmark artifacts, public outputs divide cleanly into two classes. Most rows—Mem0's `memory-benchmarks` platform, A-MEM/AgenticMemory, SNAP LoCoMo RAG, Backboard, MemoryLake, Engram, and external AMB comparison rows for Mem0, Zep, Letta, LangMem, and MemoryBank—do not commit per-question retrieved context, retrieved memory ids, or LoCoMo evidence ids in their released artifacts, and so admit no published-output PFLC replay. Three AMB LoCoMo runs do: Hindsight (1,540 cases), hybrid-search (1,540), and Cognee (152). Each injects per-question context whose source blocks carry recoverable LoCoMo `dia_id`s, enabling a context-derived PFLC replay against the official LoCoMo `qa[].evidence` field.

Table 2: Published-output PFLC on AMB LoCoMo runs (lookup-relevant denominator).

Run	Scored	Lookup-rel.	PFLC@10	PFLC@20	PFLC@50
AMB `locomo-hindsight`	1,540	1,536	65.8%	83.1%	97.6%
AMB `hybrid-search`	1,540	1,536	64.8%	75.7%	90.5%
AMB `cognee`	152	150	60.0%	73.3%	92.0%

The replayable runs are not lookup-contract collapses; they decompose LoCoMo's aggregate answer accuracy into three components that the aggregate conflates. For example, AMB hybrid-search produces 276 cases where the answer is wrong while every gold evidence `dia_id` is already present in the emitted context by PFLC@50, isolating an answering or judging residual from a retrieval-contract residual; conversely, 100 cases are correct while all gold evidence is missing at PFLC@50, flagging answer success not fully backed by the LoCoMo evidence contract as scored here. PFLC@10 and PFLC@20 carry the rank-sensitivity signal because Hindsight averages 36,235 context tokens per question.

The operational claim is therefore narrow: once per-question context is published, PFLC decomposes LoCoMo scores into evidence exposure, evidence rank or context saturation, and answer-generation residuals. Without those artifacts, the same decomposition is impossible, which is why the artifact-blocked subset remains the headline survey gap.

4.5 LongMemEval Transfer Probe

The LongMemEval transfer probe produces an inspectable negative external result. The run covers 1,935 policy/cell/case rows across `primary_contract`, `path_a_only_denominator`, and `path_b_only_denominator`. Candidate-stream hash invariants pass in all cells. The headline result is shown in Table 3.

Table 3: LongMemEval transfer probe: CQ-vs-Reflection result.

Cell	`all_hit_at_50` delta	`any_hit_at_50` delta
`primary_contract`	-1.0	0.0
`path_a_only_denominator`	-1.0	0.0
`path_b_only_denominator`	-1.0	0.0

The stable sign fires the preregistered-gate condition, but the sign is negative. CQ retrieves at least one gold evidence session in every case and therefore ties eager baselines on `any_hit_at_50`. It never retrieves both evidence sessions, so it loses `all_hit_at_50`. The methodology gate passes, but transfer is CQ-negative and isolates a policy-facing pending-readout gap.

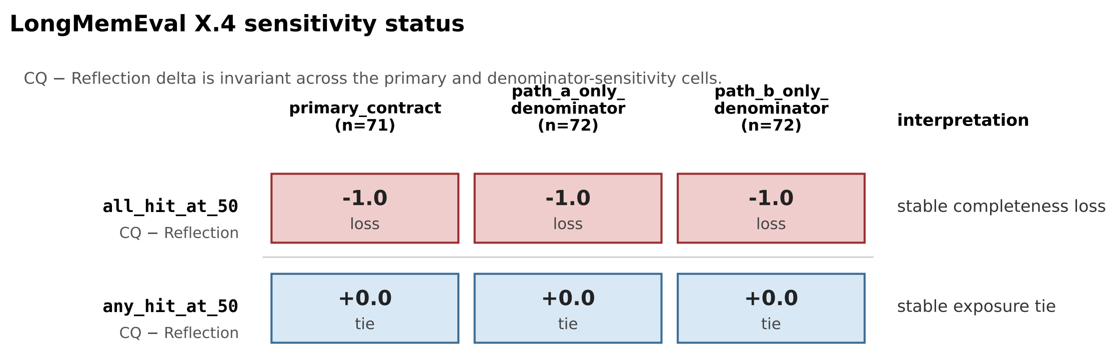

Figure 6: LongMemEval transfer probe sensitivity status. The CQ-vs-Reflection delta is stable across the primary and denominator-sensitivity cells: -1.0 on `all_hit_at_50` and 0.0 on `any_hit_at_50`.

4.6 Pending Multi-Evidence Follow-Up

The follow-up is structured around a paired pre-registration: a CQ-side *repair* hypothesis and a Reflection-side *symmetric falsifier*. The repair hypothesis is that the transfer-probe loss is induced by answer-time readout cardinality, not by CQ-only storage or asymmetric inputs. If that is correct, two things must hold in the same cells: (i) `CQPendingMultiEvidence` — which changes the no-durable pending fallback to expose all eligible same-slot pending candidates, with no change to storage, adapter, judge, or headline metric — must close the gap on `all_hit_at_50`; and (ii) `ReflectionEagerWriteCardinalityCapped` — which preserves Reflection's winning write path and only caps answer readout to one durable candidate id — must reproduce the original base-CQ loss with the same magnitude. The symmetric falsifier is the load-bearing half: it rules out CQ-only storage and asymmetric-input explanations for the loss by reproducing it on a baseline that previously won the metric.

Both predictions hold (Table 4). Plural pending readout moves base CQ from 0/71 to 71/71 in the primary cell; capping Reflection moves it from 71/71 to 0/71 in the same cell. The pattern is stable across both denominator-sensitivity cells.

Table 4: Pending multi-evidence follow-up on `all_hit_at_50`: the repair (`CQPending MultiEvidence`) and the symmetric falsifier (`ReflectionEagerWriteCardinalityCapped`).

Cell	Base CQ	Follow-up CQ	Reflection	Capped Reflection
`primary_contract`	0/71	71/71	71/71	0/71
`path_a_only_denominator`	0/72	72/72	72/72	0/72
`path_b_only_denominator`	0/72	72/72	72/72	0/72

The internal regression gate reports maximum checked overall delta 0.0 across forced contradiction, preference drift, mechanism-diverse held-out, adversarial upstream noise, and evidence-conflict spectrum. Focused negative-control tests confirm that below-floor poisoned siblings, contested siblings, wrong-scope siblings, mirrored-source weak siblings, and demoted siblings are not returned by plural pending lookup. With the symmetric falsifier holding, the supported mechanism is answer-time readout cardinality, not a richer CQ-only memory substrate; the repair does not claim broader CQ-vs-Reflection superiority.

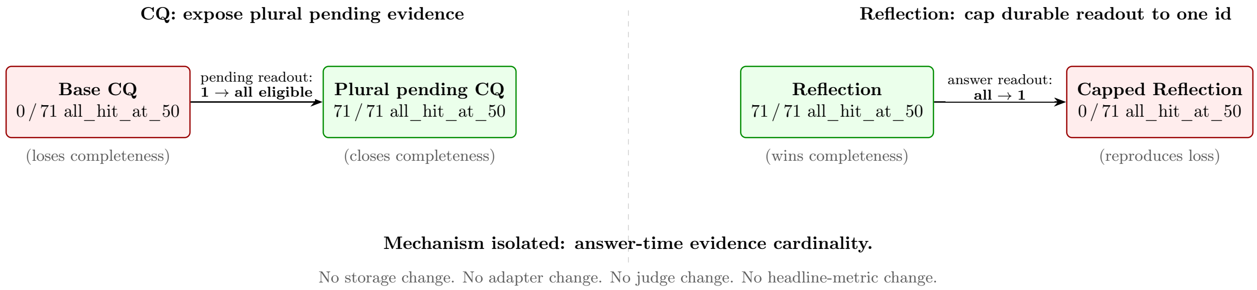

Figure 7: Cardinality-repair follow-up. Changing the CQ pending-readout cardinality from singleton to plural converts a 0/71 loss into a 71/71 win on `all_hit_at_50` in the primary cell. Capping the durable Reflection readout to a single id reproduces the original CQ loss with the same magnitude. The symmetric swap isolates answer-time evidence cardinality as the responsible interface variable; nothing else in the pipeline changes.

5 Discussion

The results map the conditions under which CQ's mechanisms remain visible. When the noisy stream preserves contradiction edges and the policy-query surface stays aligned, CQ's contestation and demotion behavior survives. If the stream or interface omits canonical slots, source identity, scope handles, or the relevant policy-facing identifier, policy ties become evidence about the benchmark surface rather than about policy equivalence.

The negative and mixed rows carry that evidence. The 7B gate shows that aggregate component metrics can hide concentrated failures. The 32B mixed-support audit leaves several ties as bounded nulls rather than convergence evidence. The lookup diagnostic shows that clustering-quality metrics can pass perfectly while exact lookup handles fail. The LongMemEval transfer shows that an external adapter can pass methodology gates and still produce a stable CQ-negative result. The pending multi-evidence follow-up then isolates one interface failure mechanism without rewriting the original result.

The main claim is methodological. Same-stream memory-policy evaluation makes policy distinctions auditable. It surfaces positive mechanism survival (forced contradiction under 32B extraction) as well as cases where a policy distinction is hidden by the interface (exact policy-query canonical-id failure under perfect clustering, or evidence completeness failure under singleton pending readout). The benchmark reports both outcomes because both affect whether a memory policy can be evaluated.

The CQ case study also illustrates why immediate-write baselines must be strong and fair. Reflection wins some rows for mechanistically uninteresting reasons, such as overwriting into a durable state that happens to include both evidence sessions, but the comparison is still valid because the baseline received the same stream and used the same substrate. The follow-up uses that interpretation directly: the interface change is registered before execution, and the capped-Reflection control serves as a preregistered mechanism falsifier — it rules out CQ-only storage and asymmetric-input explanations by reproducing the original loss on a winning baseline through a single-variable readout change. This isolates answer-time readout cardinality as the mechanism, but does not generalize to a broader metric-fitting defense.

6 Extractor Reproducibility Check

The extractor reproducibility check bridges failure-taxonomy evidence and noisy policy execution. It reruns the locked 7B anchor and two 32B primary cells under preregistered model tags, schema profiles, digest checks, and frozen sentinel requirements. The 7B anchor reproduces the locked failure counts. Both 32B cells clear the preregistered gate, with zero primary scenario errors, zero primary observed failures, zero aggregate failures, and zero frozen-sentinel observed failures.

This section is intentionally short because the unlock is not itself a memory policy result. Its role is licensing. It shows that the later noisy same-stream policy comparison was not run on an extractor stream that the preregistration would have rejected, while the 7B anchor remains a clean example of an aggregate-passing false unlock that the full gate prevented.

7 Limitations

The noisy-mode support is mechanism-local. CQ separates from `ReflectionEagerWriteLite` on forced contradiction and partially on preference drift under the current 32B artifacts, ties `Mem0WritePolicyLite` on the frozen sentinel, and leaves source-independence, scope-contamination, useful-pending, and poisoning-defense rows unsupported under the current evidence.

The LongMemEval result is also bounded. The completed transfer probe is stable and negative for base CQ on evidence completeness. The pending multi-evidence follow-up supports a readout-cardinality repair for that loss. LongMemEval-V2, LongMemEval-S, distractor-heavy retrieval quality, and natural-language QA success remain future evaluation targets; the `answer_correct` surface is diagnostic-only in the transfer probe because policy answers are trace-shaped.

PFLC is a diagnostic class rather than a new pass/fail gate. The field-level survey records what released baseline artifacts expose. LoCoMo's evidence field is a clean gold-side target, and MemoryAgentBench's conflict-resolution tasks are PFLC-relevant. The survey outcome splits: most released artifact contracts and standard output surfaces are blocked for PFLC, while the AMB LoCoMo subset admits a published-output replay that decomposes aggregate accuracy without resolving the blocked majority.

The resolution audit replay repair remains an abort (repo: Bucket D). The lookup diagnostic leaves the aborted audit and the noisy same-stream policy comparison's mixed-support verdict unchanged, and the blocked cross-tab of lookup hit versus per-policy answer success remains unrun. A third resolution- audit attempt would require either materializing the original locked run JSON payloads or preregistering a fresh policy-comparison baseline with archived payloads.

Finally, all experiments are local-first prototypes. Real-user weeks-long helpfulness, learned semantic scope inference, production retrieval stacks, and API-hosted extractors remain future evaluation targets.

8 Future Work

Closing bounded-null rows under stronger extraction. The noisy same-stream policy comparison leaves four rows—source-independence, scope-contamination, useful-pending, and poisoning—bounded by component evidence rather than by policy behavior. A preregistered extension would run these rows on a stronger extractor cell, either a larger local model or an API-hosted extractor under the same redacted same-stream guarantee. The primary prediction is that at least scope-contamination and useful-pending separate CQ from Reflection with a 95%

lower confidence bound above zero. A negative outcome re-attributes those rows from component evidence to policy ties, which is itself a stronger claim than the current bounded-null label.

LongMemEval beyond the v1 transfer probe. The pending multi-evidence repair is scoped to the v1 evidence-only split. LongMemEval-S introduces distractor sessions that can shift the dominant failure mode from completeness loss to precision loss. A preregistered extension keeps evidence-id completeness as the primary PFLC target, adds `all_hit_at_k` at small k as a precision-sensitive co-primary, and classifies the predicted failure mode before unblinding. Both capped Reflection and plural-pending CQ run in the same cells; the preregistered-gate condition (repo: Bucket A) requires a stable sign across at least two contract-sensitivity cells, matching the transfer probe criterion.

External-transfer boundary to conflict resolution. MemoryAgentBench's conflict-resolution tasks are the closest external surface to CQ's contestation and demotion mechanism. A rigorous transfer must satisfy three preconditions before policy execution: an adapter that exposes per-question identifiers sufficient for PFLC scoring; hidden-answer verification analogous to the LongMemEval dual-path protocol; and a preregistered abort criterion (repo: Bucket D) if the source benchmark does not commit to a stable evidence id across releases. A transfer that cannot meet all three is labeled descriptive-only at preregistration time, not retroactively.

Resolving the resolution audit abort. The blocked resolution audit cross-tab is the one outstanding internal lock (repo: CQR Bucket D). Two paths are acceptable. The first materializes the original locked policy-comparison run JSON payloads if recoverable. The second preregisters a fresh policy-comparison baseline that mandates payload archival and reports the lookup-versus-answer-success cross-tab as a primary readout. The second path removes the dependency on missing artifacts. Silently retiring the cross-tab without an explicit decision record is not acceptable under the same preregistration discipline that produced the rest of this paper's results.

PFLC as a benchmark output contract. The field-level PFLC survey divides into an artifact-blocked majority and a PFLC-replayable subset (the AMB LoCoMo runs); the blocked subset is what prevents a uniform diagnostic. A concrete next step is a minimal PFLC output schema—per question, the gold evidence id set and the policy-returned id set—offered as a patch to one surveyed benchmark in the blocked subset. The success criterion is adoption by at least one benchmark maintainer. The acceptable failure criterion is principled rejection on the grounds that the schema cannot represent a specific task family. Silent neglect is the failure mode this paper's survey already documents and is not acceptable evidence against the schema.

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A Failure Taxonomy

The noisy extractor-quality gate taxonomy (repo: Phase 3) groups component failures into four recurring modes: required-field omission, ID-namespace confusion, durable-claim drift into temporary/session framing, and contradiction-edge misses. These categories are used to explain why the 7B aggregate-passing run stayed locked and why some 32B rows remain unsafe to interpret as policy convergence.

Representative inspection traces are:

- `data/results/audit_trace_forced_contradiction_default.html`
- `data/results/audit_trace_scope_contamination_default.html`
- `data/results/audit_trace_useful_pending_memory_default.html`

The traces are qualitative inspection aids. Numeric claims come from metrics, manifests, component summaries, and compact audit evidence.

B Reproducibility

The repository records provenance at several layers: preregistration lock hashes, adapter pins, model tags and digests, prompt/schema profiles, candidate-stream hashes, clean-worktree checks, Python version, commands, and artifact hashes. Small headline verification artifacts are committed. Large per-scenario run payloads are regeneratable-only unless a preregistration requires byte-stable replay.

The draft was prepared on branch `paper/ship-paper-draft`, based on:

`longmemeval-externalization-gate`

The Phase Z plan allowed either merging the LongMemEval follow-up branch into `main` or pinning the paper package to the exact branch and commit. Because `main` and the pinned branch diverged after the earlier merge, this draft uses the branch-pinning path. The reproducibility README in `paper/repro/README.md` records that decision and lists the commands for local verification.

The CQR replay aborts are reproducibility evidence. The path-normalized repair preserved stable checks and still refused to emit a verdict when locked run JSON payloads could not be reproduced. The paper treats that as support for strict replay discipline.

C Preregistration Locks

The main preregistration and lock documents are:

- `docs/preregistered_memory_governance_evaluation_template.md`
- `docs/preregistration.md`
- `docs/local_unlock_probe_preregistration.md`
- `docs/noisy_policy_comparison_preregistration.md`
- `docs/adversarial_upstream_noise_preregistration.md`
- `docs/adversarial_upstream_noise_dated_followup_preregistration.md`
- `docs/abstention_quality_preregistration.md`
- `docs/canonical_id_resolution_audit_preregistration.md`
- `docs/canonical_id_resolution_audit_repair_preregistration.md`
- `docs/qr_canon_audit_registration.md`
- `docs/policy_facing_lookup_contract_registration.md`
- `docs/fair_stream_externalization_preregistration.md`
- `docs/cq_pending_multi_evidence_preregistration.md`

The pending multi-evidence follow-up lock SHA is:

`84b3c44148494c1b6f72e8201a8da886f8e1f48e49ce00197f535c7e08638840`

The canonical-id PFLC alias function remains the byte-locked CQR alias function; the PFLC workstream does not generalize that alias rule to retrieval-id, slot-id, fact-id, or answer-handle instances.

D Artifact Index

The most important artifact paths for this draft are:

- docs/predictions_vs_results.md
- docs/adversarial_upstream_noise_results.md
- docs/adversarial_upstream_noise_dated_followup_results.md
- docs/abstention_quality_results.md
- data/results/component_gate_decision_general_v1_summary.json
- data/results/component_gate_decision_qwen2_5_7b-instruct-q4_K_M_default_summary.json
- data/results/component_gate_decision_qwen2_5_32b-instruct-q4_K_M_default_summary.json
- data/results/component_gate_decision_qwen2_5_32b-instruct-q4_K_M_scenario_conditioned_summary.json
- docs/noisy_policy_comparison_results.md
- data/results/noisy_policy_comparison_summary.json
- docs/noisy_policy_mechanism_audit.md
- data/results/noisy_policy_mechanism_audit_evidence.json
- docs/canonical_id_resolution_audit_results.md
- data/results/canonical_id_resolution_audit_stop_2026-05-15.json
- data/results/canonical_id_resolution_audit_stop_2026-05-16.json
- docs/qr_canon_audit_results.md
- data/results/qr_canon_audit_metrics.csv
- docs/policy_facing_lookup_contract_proposition.md
- docs/qr_canon_field_diagnostic_results.md
- data/results/qr_canon_field_diagnostic_metrics.csv
- docs/locomo_baseline_replay_survey.md
- data/results/locomo_amb_pflc_summary.json
- docs/longmemeval_transfer_results.md
- data/external/longmemeval/transfer_summary.json
- data/external/longmemeval/transfer_per_case_rows.csv
- data/external/longmemeval/transfer_manifest.json
- docs/cq_pending_multi_evidence_results.md
- data/external/longmemeval/transfer_followup_summary.json
- data/external/longmemeval/transfer_followup_per_case_rows.csv
- data/external/longmemeval/transfer_followup_manifest.json

Figures ship in two forms. The five conceptual diagrams (CQ lifecycle, evidence ledger, externalization protocol, PFLC formal gap, cardinality-repair follow-up) are TikZ source files under `paper/figures/` that PDFLaTeX compiles inline at build time. The two quantitative charts (noisy same-stream policy comparison mechanism audit; LongMemEval transfer probe cells) are matplotlib output committed as paired PNG/PDF and regenerated from the committed artifacts above by `paper/repro/render_charts.py`. All figure values trace to the documents and result JSONs listed in this index; the AMB LoCoMo PFLC numbers in Section 4.4 come from `data/results/locomo_amb_pflc_summary.json` (SHA recorded in the summary file's manifest).